Decompose a Goal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Decompose a Goal before opening notes. Show any later correction separately.

Try the re**call** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Decompose a Goal trace and record the first step you can explain without help.

Use W2: Follow the execution order for **Function**s and **argument**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Decompose a Goal explanation against the supplied facts before discussing it.

Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Create a new function independently

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Decompose a Goal target to a trace and a justified conclusion.

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Decompose a Goal case and show the deciding evidence.

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Decompose a Goal adds to Transparent Book Recommender and keep the evidence.

Separate reader profile, book data, score calculation, ranking, and explanation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Label the parts

> Label the parts. Apply the supplied Decompose a Goal facts and keep your reasoning.

In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Follow the execution order

> Follow the execution order. Apply the supplied Decompose a Goal facts and keep your reasoning.

Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair ignored arguments

> Repair ignored arguments. Apply the supplied Decompose a Goal facts and keep your reasoning.

The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Create a new function independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Create a new function independently. Apply the supplied Decompose a Goal facts and keep your reasoning.

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 State the input assumption

> State the input assumption. Apply the supplied Decompose a Goal facts and keep your

> reasoning.

The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Decompose a Goal

> A fictional library computes total spine labels needed for several equal shelves. Explain the labelled step before checking the changed condition.

A fictional library computes total spine labels needed for several equal shelves.

Define labels(shelves,per_shelf): return shelves*per_shelf.

Call needed=labels(4,6), then print(needed). No other side effects are defined.

AI ACADEMY / AI-2/2

Decompose a Goal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Call with arguments

The call supplies shelves=4 and per_shelf=6 locally.



2. Function body

Multiply 4 x 6 to calculate 24.



3. Return value

return sends 24 back to the caller; it does not print.



4. Caller prints

needed receives 24; print(needed) displays 24.

CHANGE ONE THING

Call labels(4,0) instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What happens visibly if the **call**er stores needed but omits print(needed)?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library computes total spine labels needed for several equal shelves.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? needed receives 24; print(needed) displays 24.

1. **Function** body

2. **Call**er prints

3. **Call** with **argument**s

4. Return value

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Call**er prints needed receives 24; print(needed) displays 24. A **function** separates one named job from the surrounding program.

Arguments supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: **Call** labels(4,0) instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The returned result is 0, and the **call**er prints 0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The returned result is 0, and

the **call**er prints 0. Four shelves times zero labels per shelf is zero.

Repair a missing building block

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

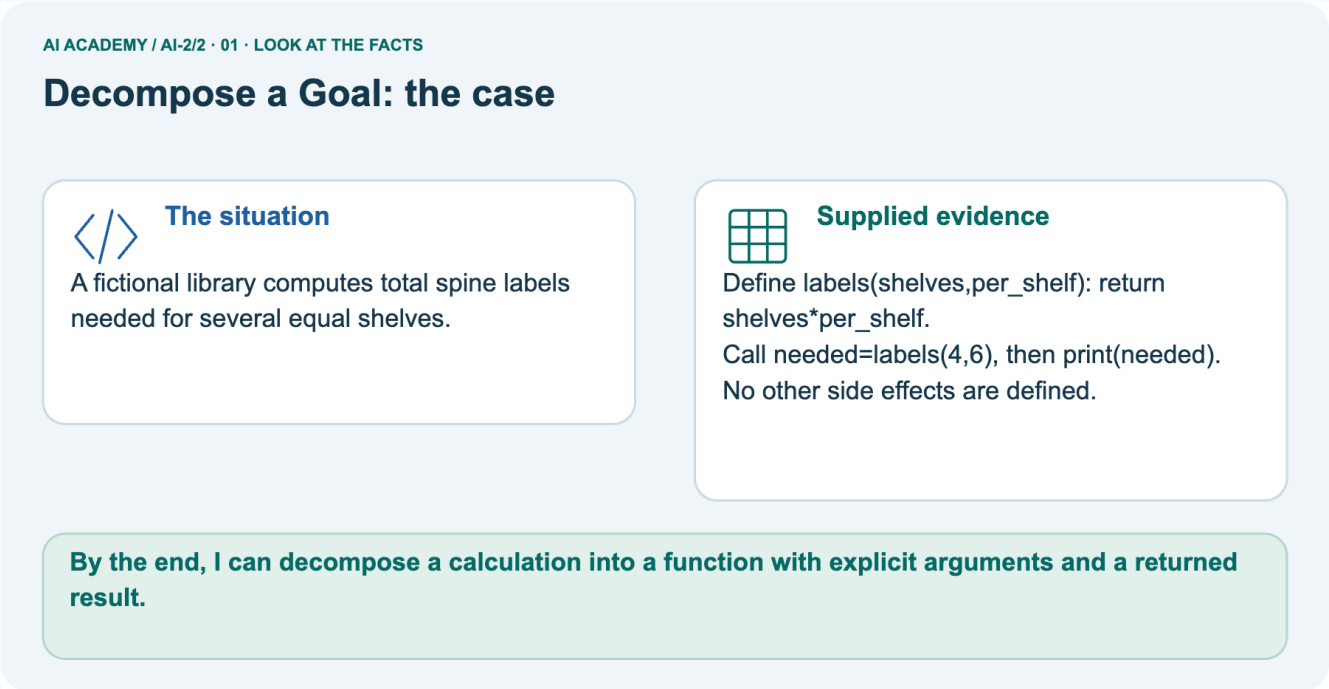
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Decompose a Goal picture from its labels, then record a relevant limit.

01. Decompose a Goal: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

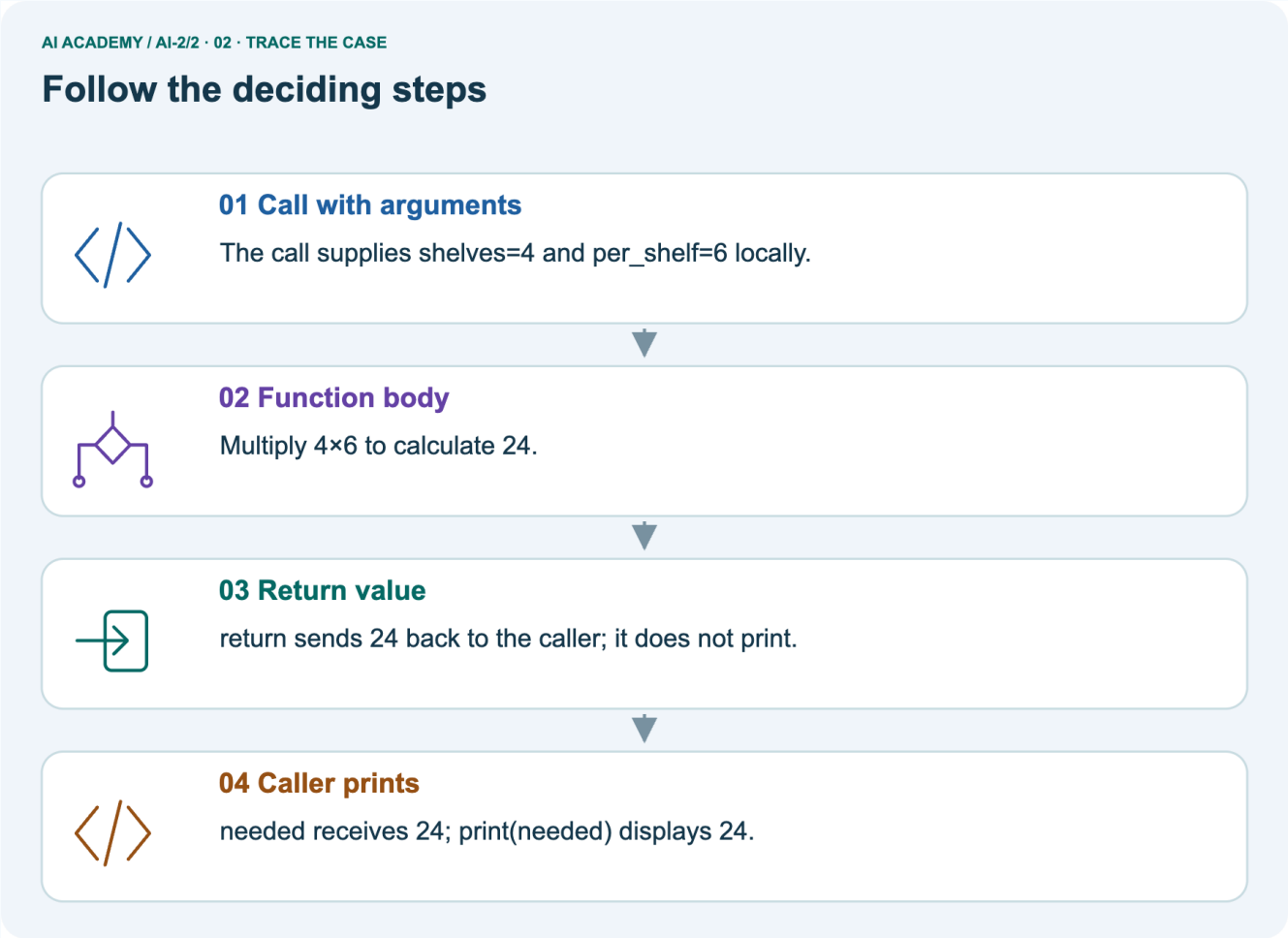
A fictional library computes total spine labels needed for several equal shelves. Supplied facts: Define labels(shelves,per_shelf): return shelves*per_shelf. **Call** needed=labels(4,6), then print(needed).

No other side effects are defined.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



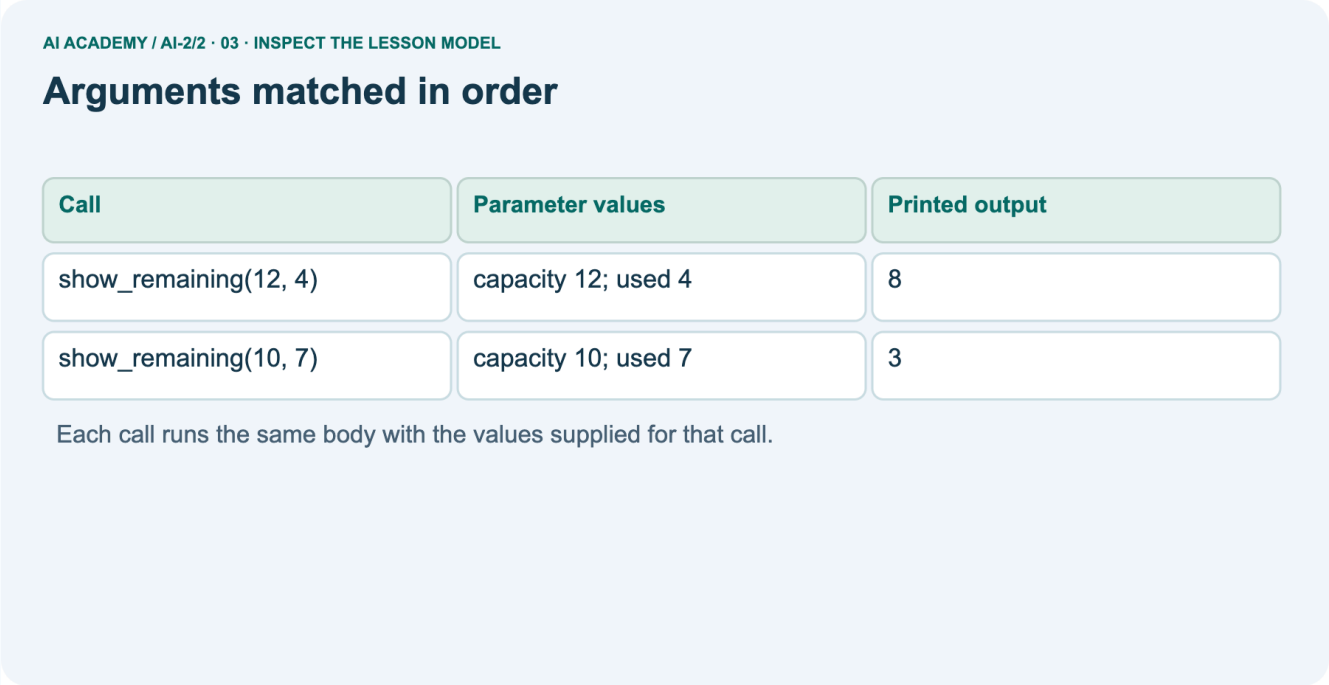
Follow the labelled picture with the matching explanation. Open the vector diagram

Call with **argument**s: The **call** supplies shelves=4 and per_shelf=6 lo**call**y. → **Function** body: Multiply 4×6 to calculate 24. → Return value: return sends 24 back to the **call**er; it does not print. → **Call**er prints: needed receives 24; print(needed) displays 24.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Arguments matched in order



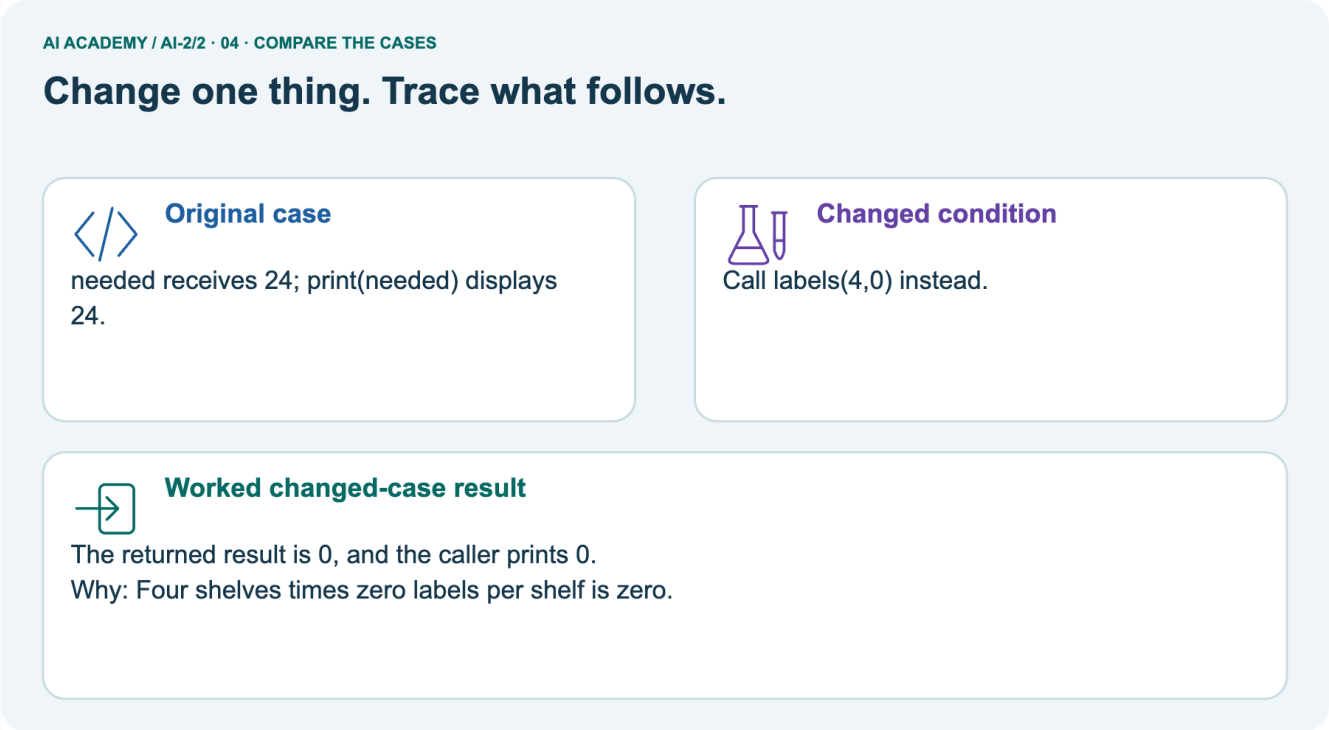
Follow the labelled picture with the matching explanation. Open the vector diagram

Arguments matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

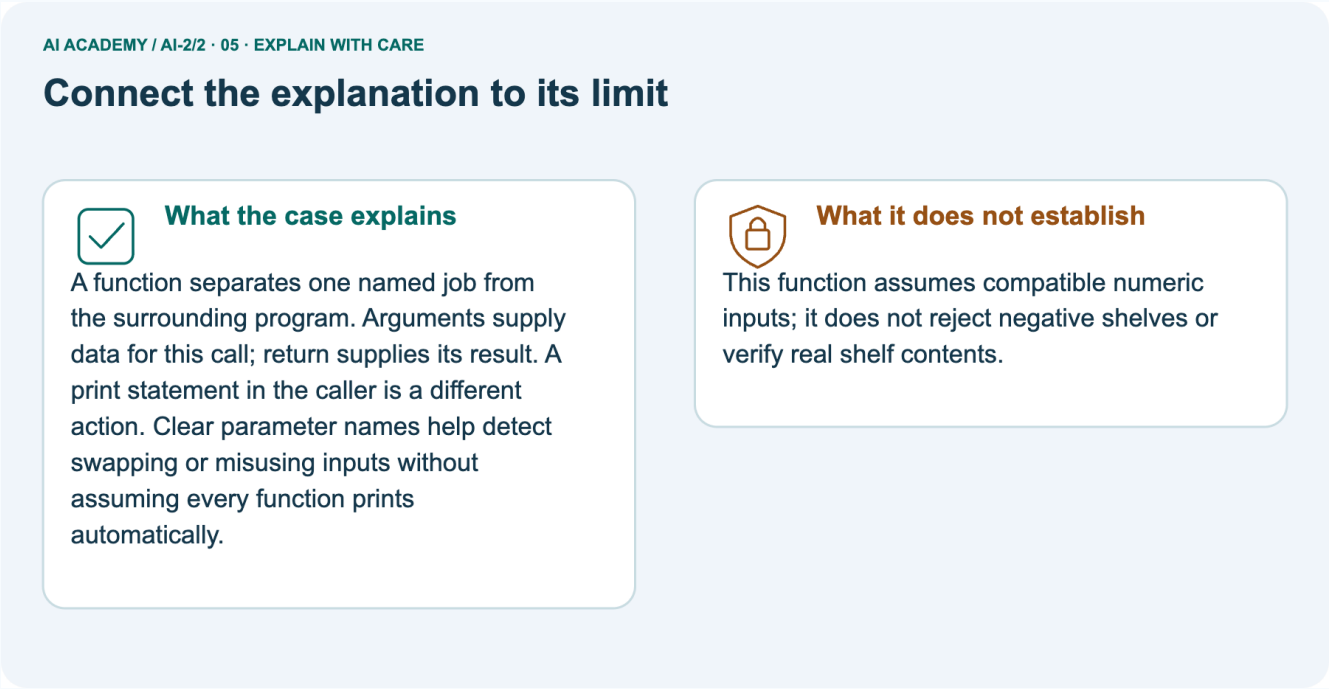
Original case: needed receives 24; print(needed) displays 24. Changed condition: **Call** labels(4,0) instead. Expected result: The returned result is 0, and the **call**er prints 0.

Reason: Four shelves times zero labels per shelf is zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

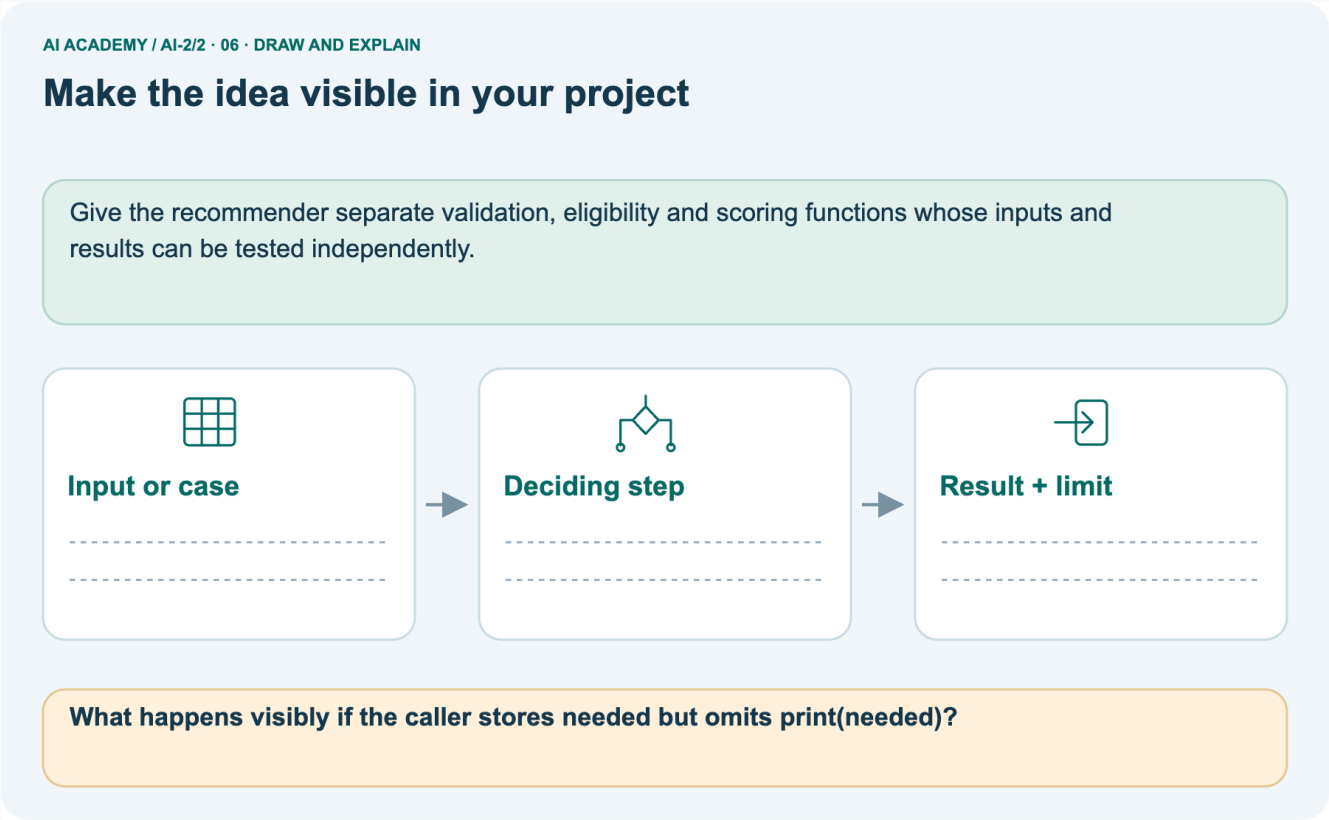
Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Limit: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently. Draw your **input**, deciding step, result and limit.

Supported question: What happens visibly if the **call**er stores needed but omits print(needed)?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.